INSTITUTE
OF LAW,
POLITICS AND
DEVELOPMENT



Sant'Anna
School of Advanced Studies – Pisa

Quantitative Methods

Module II, Session 3: Linear Regression

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1. Where we are

1.1 Last session in one slide

Standard error

How much an estimate would bounce around **if we redrew the sample**.

→ the basis for intervals (estimate ± about two standard errors) and tests (how many standard errors from zero?).

What it does not cover

Bias, measurement error, an unfair comparison.

→ Signers of Article 98 deals voted with the US *less*. The test was fine; the comparison was not.

Today we replace "the average of one group against another" with **a line**. Everything from last week carries over: every slope comes with a standard error, an interval and a test.

1.2 Quick check

Answer before the next slide — one sentence each.

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3. Why is "$p = 0.049$" not a different finding from "$p = 0.051$"?
4. In Lab A, why can the simulated spread at $n = 40$ come out *smaller* than $\text{sd}/\sqrt{n}$ without anything being wrong?

2. Reading the agent's code

2.1 Lab A, annotated

Five of you sent me your Lab A scripts. I annotated each one, line by line: what every line does, what to check, and what was done well.

On the course GitHub (download, then open in your browser): `lab_se_`*Name*`_annotated.html` for Ikram, Ksenia, Matilde, Mohammad and Samuele

- **Same prompt, different scripts, different numbers.** Three drew states *without* replacement: at $n = 40$ the spread is 3.27, below $\text{sd}/\sqrt{n} = 3.67$. Two drew *with* replacement and land on the formula (e.g. 3.76 against 3.66). Neither is a bug: 40 of 190 states uses up a fifth of the population.

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- **Comments that contradict the code.** The agent describes loops that do not exist and colours that are not in the plot.
- **Who was dropped?** Most files report *how many* rows are missing, not *which*. 26 non-states are coded 0 on `article98`, so they count as "did not sign".

2.2 A new `AGENTS.md`

The scripts were far more complicated than the task. The new `AGENTS.md` tells the agent to:

- write no functions of its own, and never loop over variables or settings, or store results in lists

It ends with two complete example scripts: one to imitate (`lab_se_simple.R`), one to avoid (the first `lab_se.R`).

Now: download it from the course GitHub, replace the one in your project folder, and type `/new` so OpenCode reads it.

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- keep output, plots and comments plain and short; use simple relative paths
- print row counts, and say **which** cases it drops

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3. From averages to lines

3.1 Today's question

Goldsmith & Horiuchi (2012): does what foreign publics think of US foreign policy shape how their governments act towards the US?

- **Opinion about US foreign policy**: share saying US policy has a *positive* effect on their country, minus share saying *negative*. Gallup, 2002–03, 58 countries. Runs from −1 to +1.

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- **UN voting with US**: percentage of the 2003 General Assembly votes the US called "important" on which the country voted as the US did.
- Today we replicate the two **regression** models for UN voting (Table 3, Models 5 and 6). The other four models are for Session 4.

3.2 Lab A: specify, run, verify

To the agent

Write scripts/lab_ols.R. Using data_2011-07-26.dta, replicate Models 5 and 6 of Table 3 in Goldsmith & Horiuchi (2012): OLS of unvoting on opinion about US foreign policy, first alone, then with the eleven controls. Keep the 58 countries with keep == 1. Build opinion and the controls as the paper describes them (pp. 563, 574–75). Use the paper's robust standard errors. Print ours next to the published ones. Run it.

Verify — without asking the agent:

1. Model 5 should give 2.07 [9.06], Model 6 15.78 [3.62]. If not, check every control against the paper's definition *and* against `summary()` of the variable.
2. How many countries per model? `lm()` drops incomplete rows *silently*: check `nobs()`.
3. Did the agent call 15.78 an effect, or interpret a control? One line in `log.md`.

4. The story thus far

4.1 Today's question as a table

Split the 58 countries in two, twice: by **opinion** (more positive than negative views of US policy, or not) and by **UN voting** (above or below the median, 58%).

	Voting congruence with US		
	High	**Low**	**Total**
Favourable opinion of US	8	9	17
Unfavourable opinion of US	19	22	41

- About half of each group votes "high". Average UN voting: 52% (favourable) against 51.6% (unfavourable).

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- About half of each group votes "high". Average UN voting: 52% (favourable) against 51.6% (unfavourable).
- Does that mean opinion makes **no difference**? Only if the two groups are comparable.

4.2 What we see, and what we do not

Each country has **two** potential outcomes: how it would vote if its public were favourable (Y_{fav}), and if it were unfavourable (Y_{unfav}). We only ever see one.

	Y_{fav} if public favourable	Y_{unfav} if public unfavourable
Favourable ($Fav = 1$)	observed: 52	? counterfactual
Unfavourable ($Fav = 0$)	? counterfactual	observed: 51.6

- The effect we want: for the **same** countries, Y_{fav} minus Y_{unfav}.

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- What we can compute: the favourable countries' Y_{fav} minus the **other** countries' Y_{unfav}.
- The red cells are what we would need to know, and never can.

4.3 The identification assumption

We may fill each red cell with the observed cell **in the same column** only if

$$E(Y_{fav} \mid Fav = 0) \approx E(Y_{fav} \mid Fav = 1)$$

and

$$E(Y_{unfav} \mid Fav = 1) \approx E(Y_{unfav} \mid Fav = 0)$$

- Unfavourable countries, **if** their publics turned favourable, would vote like today's favourable countries, **and**

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- The assumption is about the red cells: the data alone cannot check it.

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Goldsmith & Horiuchi did not assign opinion to countries. So what makes the comparison fair?

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- **Controls.** Among countries that are alike on eleven variables (income, trade, aid, democracy, NATO, …), opinion is assumed to be *as good as random*.
- So the assumption becomes: **nothing left out** that drives both opinion and voting. That is what we test today.

4.5 From categories to continuous values

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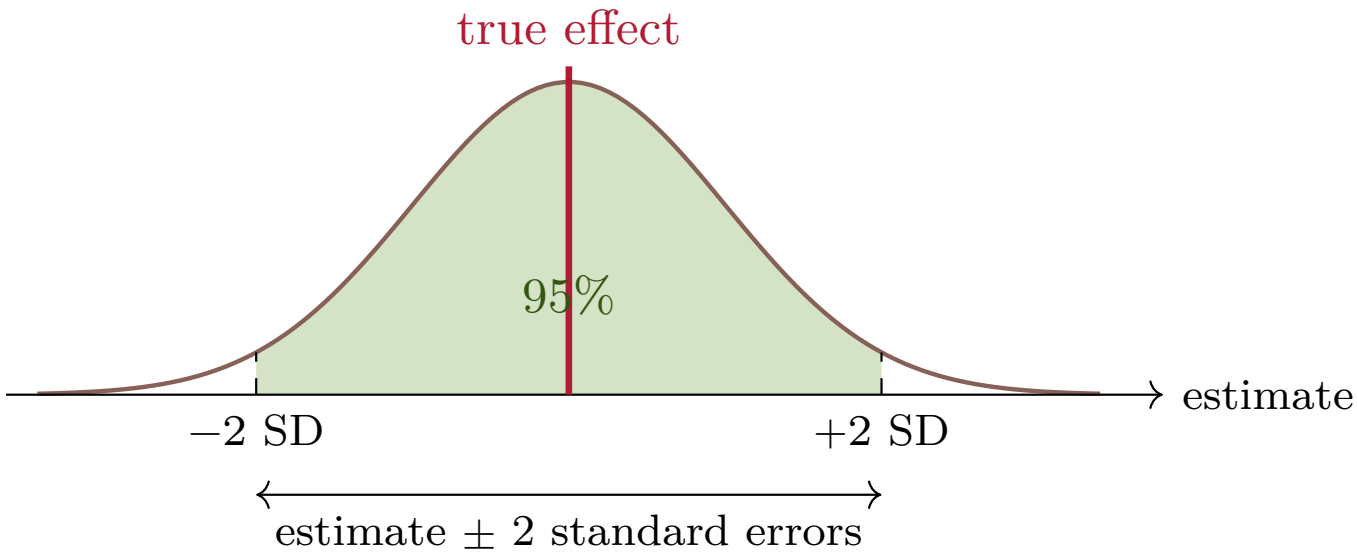
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- The logic is the same. The table now has a column for every value of opinion, and every column has red cells.
- The assumption is the same too, for any two values: countries at one level of opinion would vote like countries at another level, **if** they had that level of opinion.

4.6 Recall: the central limit theorem

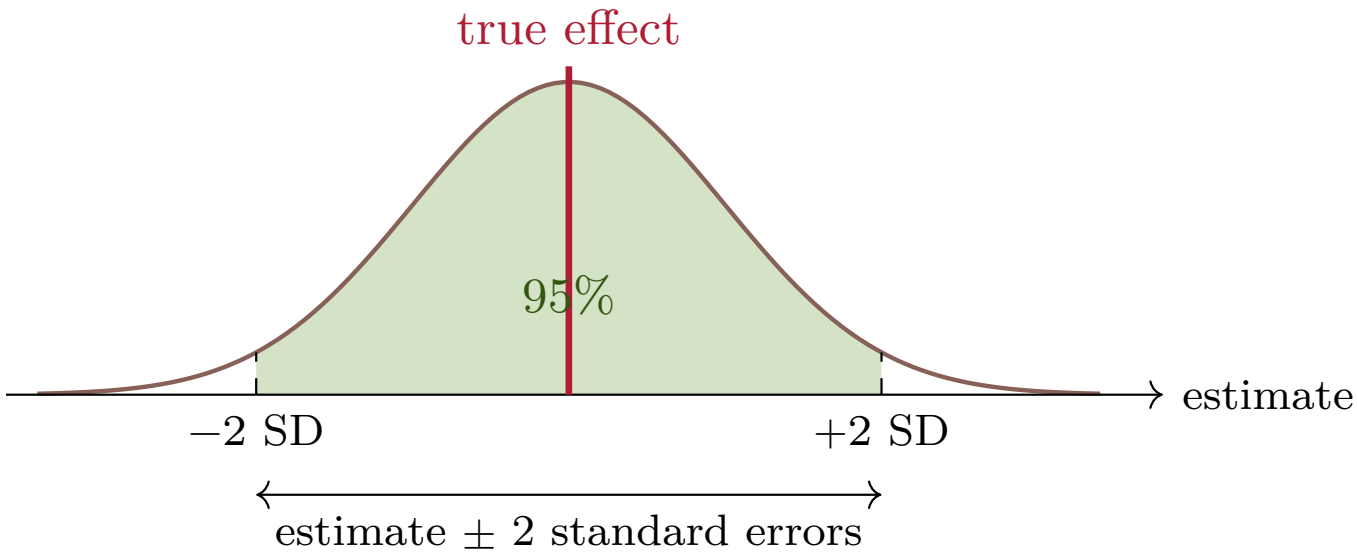
Identification tells us **where** our estimate is centred. Inference tells us **how far** it can be from the centre by chance.



- Redraw the sample many times: the estimates pile up in a **normal** curve around the true effect.

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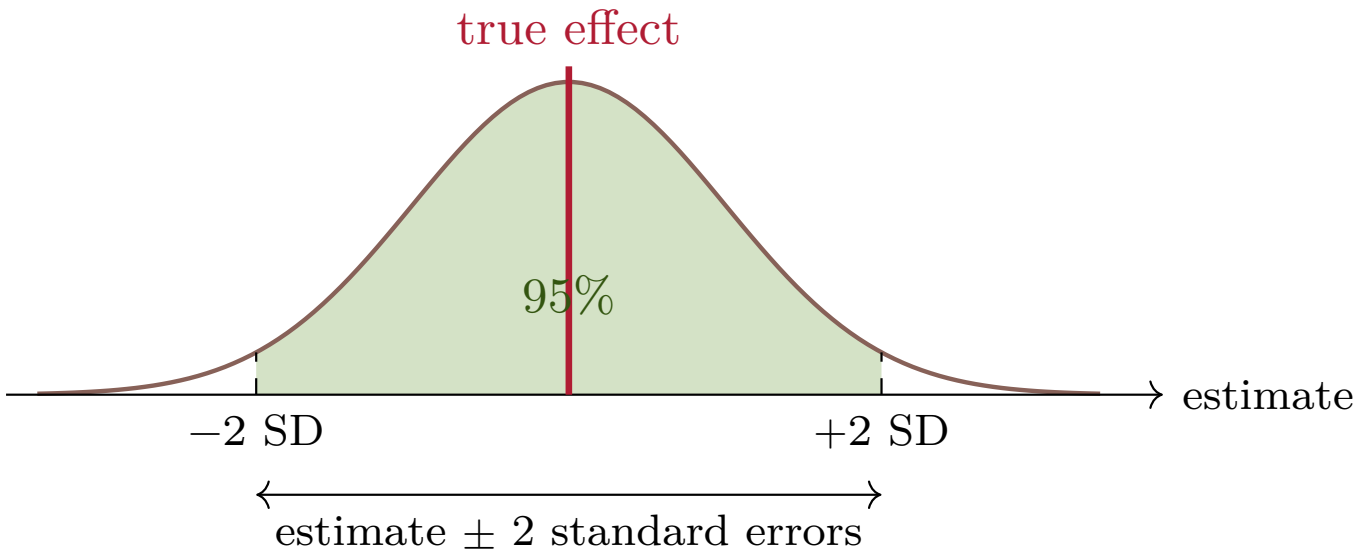
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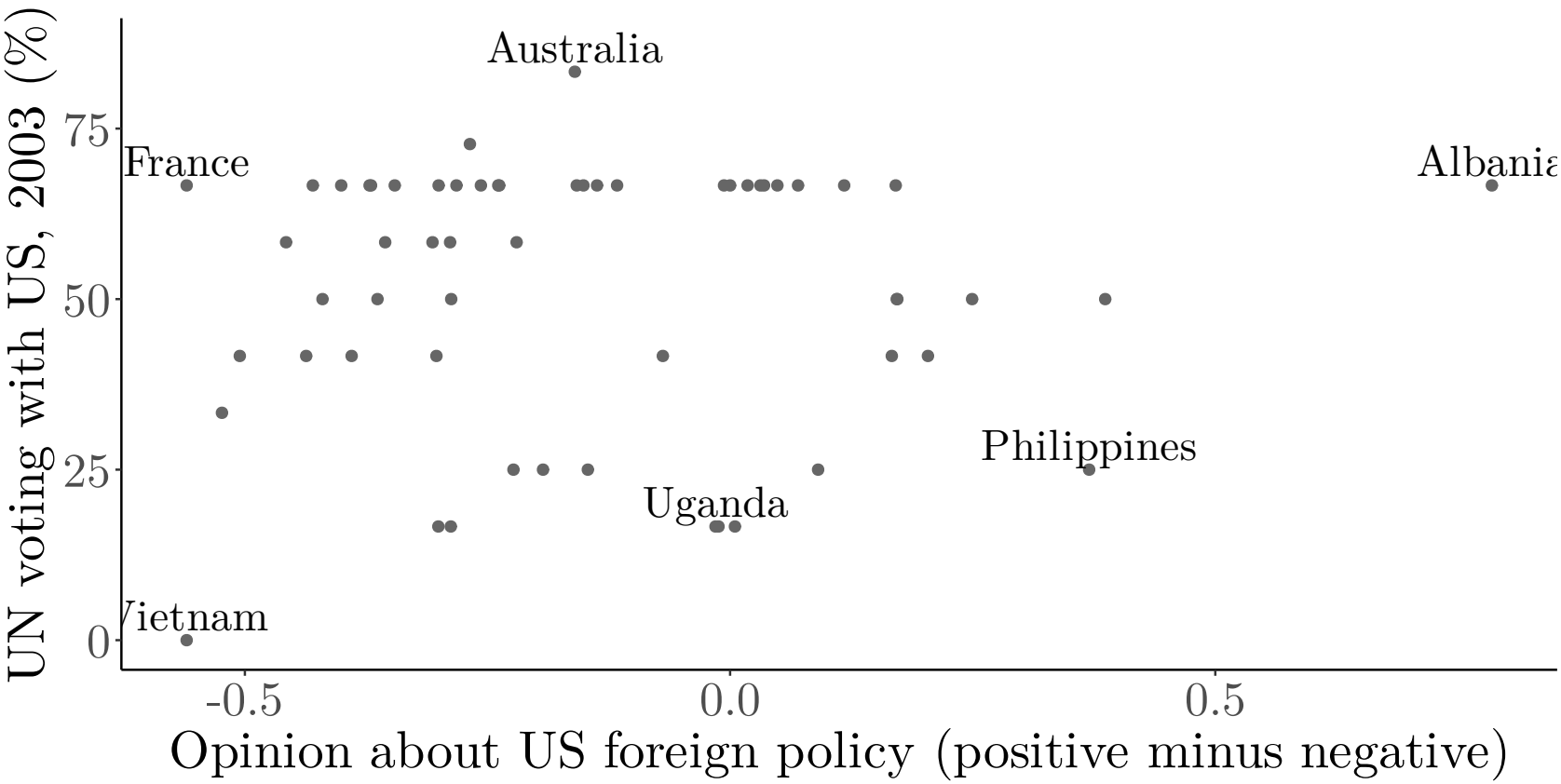
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- Redraw the sample many times: the estimates pile up in a **normal** curve around the true effect.
- Its spread (SD) is the **standard error**. About 95% of estimates fall within ±2 standard errors of the truth.
- If identification fails, the curve sits on the **wrong** value. No standard error shows that.

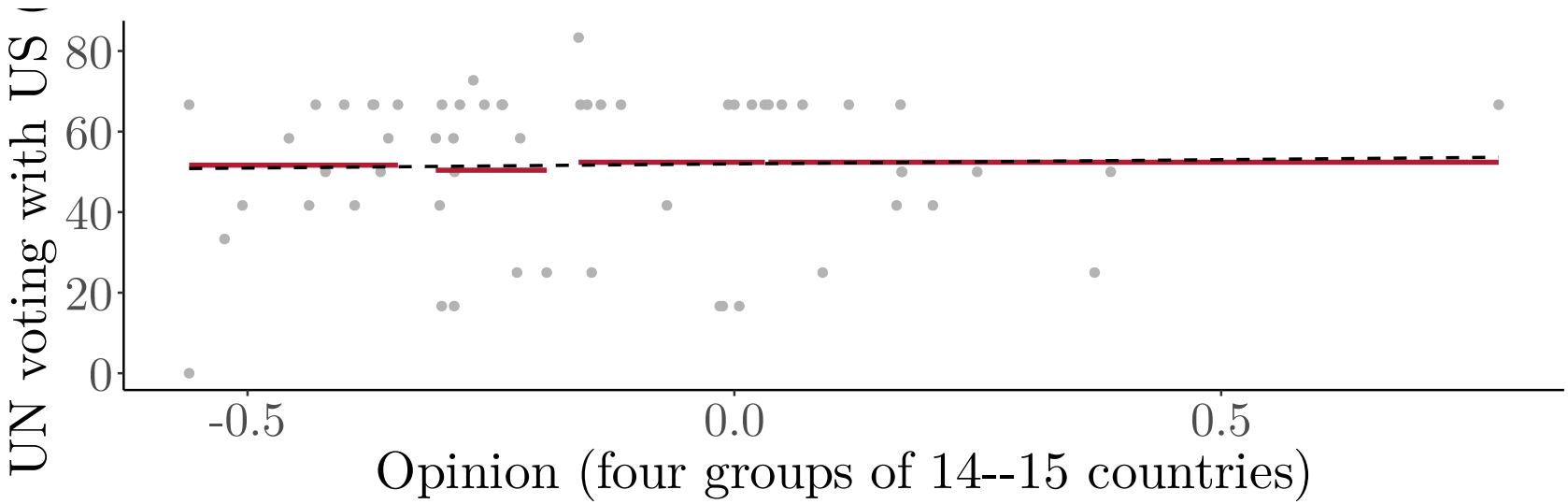
5. Drawing the line

5.1 A first look



5.2 Averages, one slice at a time

Regression starts from a question we already know how to answer: **what is the average outcome for cases with a given value of X?**



Red: the average in each quarter of the data. Dashed: the regression line — one straight line that tracks those averages as closely as it can.

5.3 What the line is

- Regression predicts the **average** outcome for each value of X. It is a summary of conditional averages, not a description of any single country.

Two facts that hold for every OLS line: it passes through the point (average X, average Y), and its residuals average exactly zero (Imai 2017, ch. 4).

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- Regression predicts the **average** outcome for each value of X. It is a summary of conditional averages, not a description of any single country.
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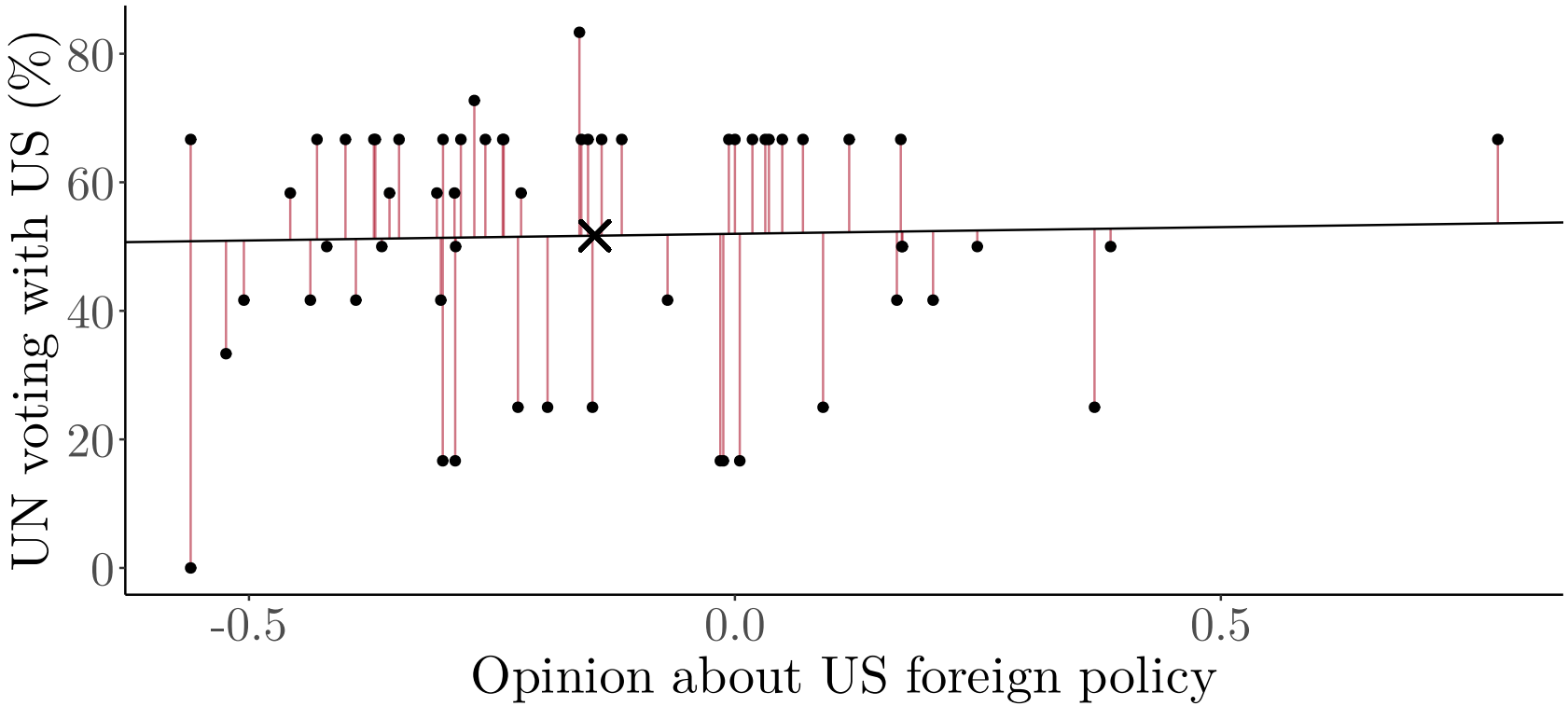
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- A straight line needs two numbers. The **intercept**: the predicted outcome where $X = 0$. The **slope**: how much the predicted outcome changes when X goes up by one unit.
- For each case, the gap between what it did and what the line predicts is its **residual**. Least squares (**OLS**, "ordinary least squares") picks the line that makes the squared residuals, added up, as small as possible.

Two facts that hold for every OLS line: it passes through the point (average X, average Y), and its residuals average exactly zero (Imai 2017, ch. 4).

5.4 Residuals, drawn



The cross marks the average of X and the average of Y.

5.5 The slope by hand

OLS has a formula. Four steps, using the cross on the last slide as the centre:

1. For each country, its **distance from the average** opinion $(x - \bar{x})$ and from the average voting $(y - \bar{y})$.

$$\text{slope} = \frac{\sum(x - \bar{x})(y - \bar{y})}{\sum(x - \bar{x})^2} \qquad \text{intercept} = \bar{y} - \text{slope} \times \bar{x}$$

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3. **Add up** the products over all 58 countries. Mostly positive: the cloud tilts up. Mostly negative: it tilts down.

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3. **Add up** the products over all 58 countries. Mostly positive: the cloud tilts up. Mostly negative: it tilts down.
4. **Divide** by the sum of the squared opinion distances, $(x - \bar{x})^2$. This turns the total into "points of voting per unit of opinion".

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5.6 Four countries' contributions

Average opinion $\bar{x} = -0.14$, average voting $\bar{y} = 51.7$.

Country	**opinion**	**voting**	$x - \bar{x}$	$y - \bar{y}$	**product**
France	-0.56	66.7	-0.42	15.0	-6.2
Albania	0.79	66.7	0.93	15.0	13.9
Vietnam	-0.56	0.0	-0.42	-51.7	21.5
Philippines	0.37	25.0	0.51	-26.7	-13.7

- Albania: far above average on both, a big **positive** product. It pulls the slope up.

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- Vietnam, at the bottom left, also pulls the slope up; France and the Philippines, above on one and below on the other, pull it down.
- Over all 58 countries the products nearly cancel out, so the slope is small: 2.07.

5.7 By hand, in R

```
x_bar <- mean(s$opinion)
y_bar <- mean(s$unvoting)
dx <- s$opinion - x_bar     # distance from mean
dy <- s$unvoting - y_bar    # distance from mean
slope <- sum(dx * dy) / sum(dx^2)
intercept <- y_bar - slope * x_bar
c(intercept, slope)
```

```
[1] 51.98386  2.07301
```

Compare with `coef(m5)` on the next slide: the same two numbers. `lm()` does nothing more than this; it just does it for many variables at once.

5.8 In R

```
m5 <- lm(unvoting ~ opinion, data = s)
coef(m5)
```

```
(Intercept)     opinion
   51.98386     2.07301
```

- **Intercept, 52:** a country whose public is evenly split (as many positive as negative) is predicted to vote with the US on 52% of important votes.

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- **Intercept, 52:** a country whose public is evenly split (as many positive as negative) is predicted to vote with the US on 52% of important votes.
- **Slope, 2.07:** one unit more opinion, 2.1 points more voting with the US.

5.8 In R

```r
m5 <- lm(unvoting ~ opinion, data = s)
coef(m5)
```

```
(Intercept)     opinion
   51.98386     2.07301
```

- **Intercept, 52:** a country whose public is evenly split (as many positive as negative) is predicted to vote with the US on 52% of important votes.
- **Slope, 2.07:** one unit more opinion, 2.1 points more voting with the US.
- **But what is "one unit"?** Going from -0.5 to $+0.5$: from France to beyond the Philippines. Nearly the whole range of the data.

5.9 Units, units, units

Before you read any coefficient, answer three questions:

1. **What is Y measured in?** Here: percentage points of votes. (In Tables 1–2 it is a share from 0 to 1. Same variable, numbers 100 times smaller.)

The same relationship can be written as a slope of 2.07 or of 0.02 depending on units. An agent will report whichever number R prints. Deciding what change is meaningful is your job.

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1. **What is Y measured in?** Here: percentage points of votes. (In Tables 1–2 it is a share from 0 to 1. Same variable, numbers 100 times smaller.)
2. **What is a one-unit change in X?** Here: the whole spread of world opinion. A **standard deviation** of opinion is 0.27.
3. **Is that change realistic?** One standard deviation more opinion goes with 0.6 points more voting with the US: a tenth of one of the 14 votes.

The same relationship can be written as a slope of 2.07 or of 0.02 depending on units. An agent will report whichever number R prints. Deciding what change is meaningful is your job.

6. The same questions, asked of a slope

6.1 A slope has a standard error

Redraw the 58 countries and the line would move. Everything from Session 2 applies:

```
round(coef(summary(m5)), 2)
```

```
            Estimate Std. Error t value Pr(>|t|)
(Intercept)    51.98       2.87   18.08     0.00
opinion         2.07       9.55    0.22     0.83
```

```
round(confint(m5), 1)
```

```
            2.5 % 97.5 %
(Intercept)  46.2   57.7
opinion     -17.1   21.2
```

Estimate, standard error, the estimate counted in standard errors (`t value`), p-value — and an interval for the slope.

6.2 What would you conclude?

- The slope is 2.1; the interval runs from -17 to 21.

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- No: the data are compatible with a sizeable *negative* relationship, with none, and with a sizeable *positive* one. With 58 countries, this regression is **uninformative**, not a finding of "no effect" (remember the STAR experiment).
- And even a tight interval would only tell us about **chance**. Not yet about whether the comparison is fair.

6.3 Robust standard errors

The paper reports "robust standard errors in brackets". What does that mean?

- The usual standard error assumes the scatter around the line is **equally wide everywhere**. Often it is not.

```
coeftest(m5, vcov = vcovHC(m5, type = "HC1"))[2, 1:2]
```

```
  Estimate Std. Error
  2.073010   9.061459
```

Exactly Model 5 in Table 3. Packages: `sandwich`, `lmtest`.

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- **Robust** standard errors drop that assumption.
- They do *not* fix bias, a wrong comparison, or cases that depend on each other.

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  2.073010   9.061459
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6.4 How well does the line fit?

Two numbers summarise how far the cases lie from the line:

- **RMSE** (root mean squared error): the typical size of a residual, in the units of Y. Here 19.2 points: the line misses by about a fifth of all votes.

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- R^2: the share of the variation in Y that the line accounts for. Here 0.001: nothing.
- A high R^2 is not the goal. A model can predict well and estimate the slope we care about badly, or predict badly and estimate it well. Next: a model with $R^2 = 0.86$.

7. Holding other things constant

7.1 Model 6: eleven controls

The paper's second model adds eleven variables that might drive both opinion and voting: troops in Afghanistan, ICC membership, alliances, NATO, US military and economic aid, trade with the US, GDP per capita, democracy, Muslim population, Western Europe.

```
m6 <- lm(unvoting ~ opinion + afgh + icc + s_lead +
           nato + aid_m + aid_e + lntrade + lngdppc +
           democ + muslimpct + europe, data = s)
round(rob(m6)[2, ], 3)
```

```
  Estimate Std. Error    t value   Pr(>|t|)
    15.784      3.622      4.358      0.000
```

From 2.07 to 15.78, with a robust standard error of 3.62. R^2 jumps to 0.86. What happened?

7.2 What "holding constant" means

A multiple regression coefficient can be computed in three steps:

1. Predict **opinion** from the eleven controls. Keep what is left over: the part of opinion the controls *cannot* account for.

So the coefficient compares countries that are **more favourable than their GDP, trade, aid, alliances ... would predict** with those that are less favourable. The controls account for 34% of the variation in opinion; the coefficient runs on the remaining 66%.

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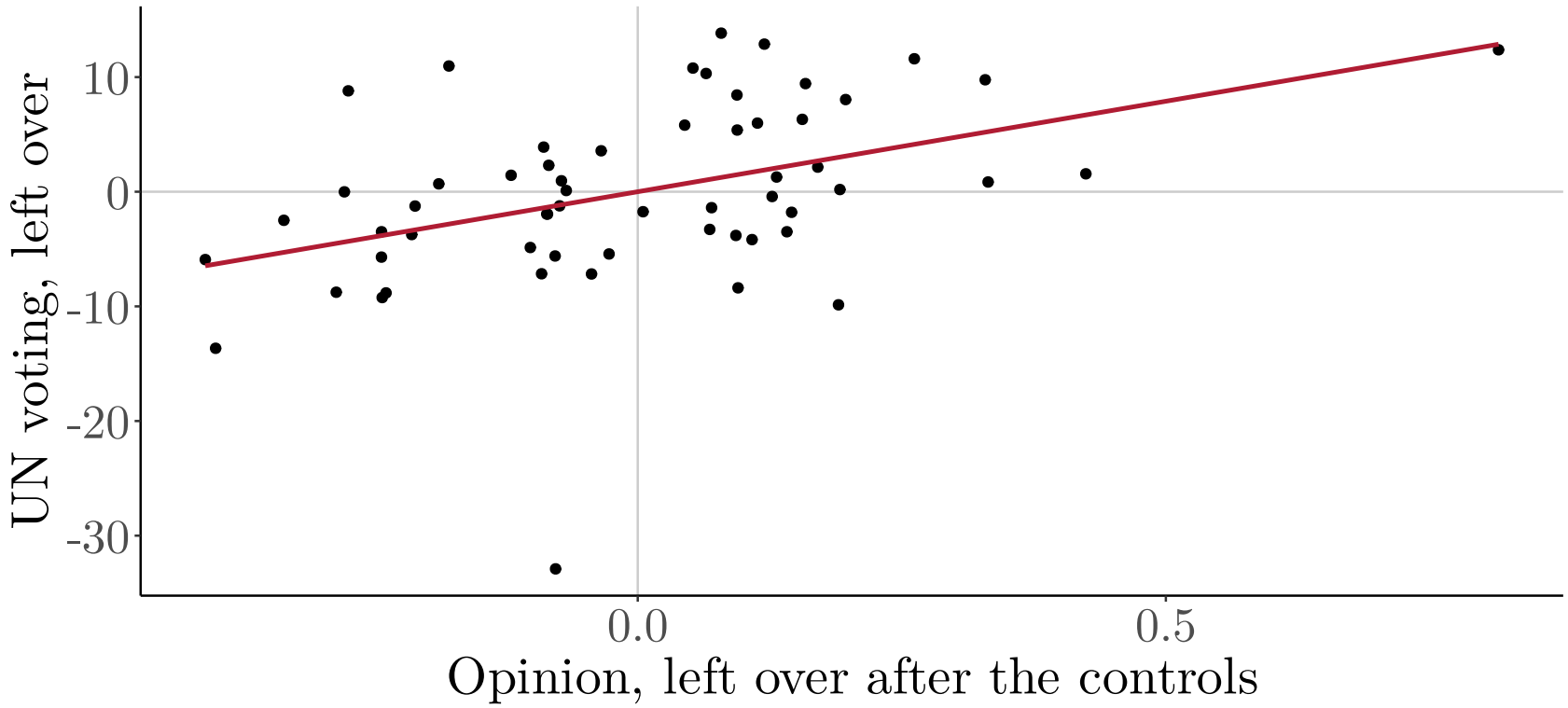
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1. Predict **opinion** from the eleven controls. Keep what is left over: the part of opinion the controls *cannot* account for.
2. Predict **UN voting** from the same controls. Keep what is left over.
3. Draw a line through leftover against leftover. Its slope is **exactly** the opinion coefficient in Model 6.

So the coefficient compares countries that are **more favourable than their GDP, trade, aid, alliances ... would predict** with those that are less favourable. The controls account for 34% of the variation in opinion; the coefficient runs on the remaining 66%.

7.3 Leftover against leftover

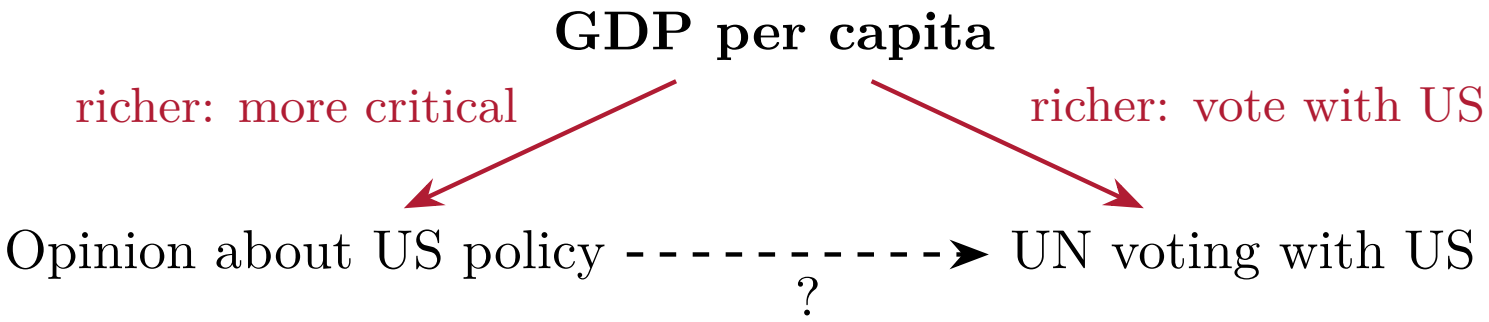


Slope: 15.78. The same number as in Model 6, from a picture you can inspect.

7.4 Why the number moved: omitted variables

Leaving out a variable biases the slope when **both** of these hold:

1. It is related to the **outcome**, holding X constant;

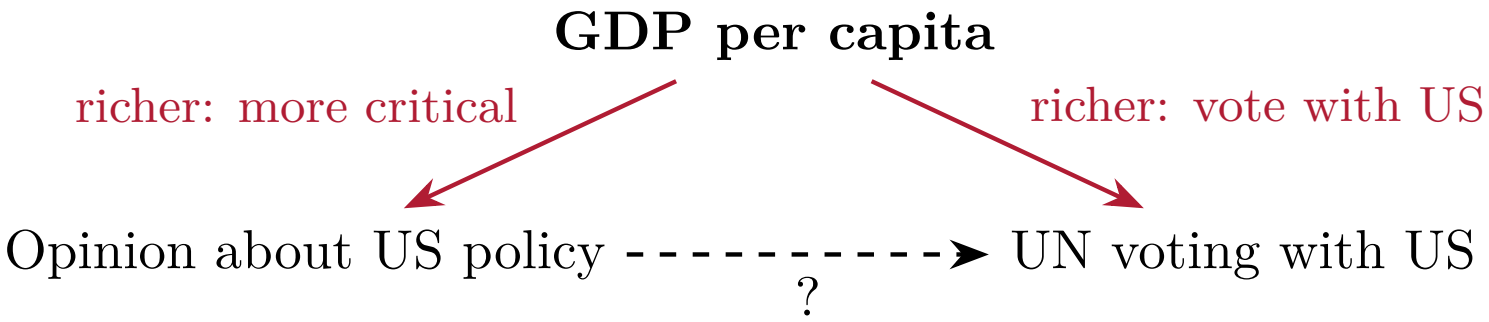


Richer countries are more critical of US policy *and* vote with the US more. Leave GDP out, and opinion gets blamed for some of what wealth does.

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2. It is related to X.



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7.5 Which way does the bias go?

	Omitted variable raises Y	**Omitted variable lowers Y**
...and goes with high X	slope too **high**	slope too **low**
...and goes with low X	slope too **low**	slope too **high**

- GDP raises voting with the US and goes with *low* opinion: the slope without it is too low. That is what we saw.

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- GDP raises voting with the US and goes with *low* opinion: the slope without it is too low. That is what we saw.
- The rule works for variables you **have not measured** too. That is its real use: ask a referee's question of any regression — *what is missing, and which way would it push?*

7.6 So: control for everything?

No. Three reasons for caution:

- **Controls caused by X.** If opinion shapes a control, adjusting for it removes part of the very relationship we want. Goldsmith & Horiuchi say so themselves (fn. 37). Troops in Afghanistan in 2002 overlaps with the first opinion survey (June–Sept. 2002): cause of opinion, or consequence?

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- **What you cannot measure stays in.** The authors' own worry: leaders who already planned to side with the US may have worked on their publics first. No control in the table captures that.
- **Few cases, many controls.** Twelve slopes and an intercept from 58 countries. Each control eats information; the estimates get noisy and sensitive to single cases.

7.7 Never read the controls

```
round(rob(m6)[c("lngdppc", "lntrade", "aid_e"), 1:2], 2)
```

```
         Estimate Std. Error
lngdppc     12.59       2.12
lntrade     -4.77       1.36
aid_e        0.46       0.12
```

- Tempting: "a 1% richer country votes 0.13 points more with the US." Don't.

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- The model was built to make the **opinion** comparison fair. Nobody asked whether it makes the GDP comparison fair: what drives GDP, and what does GDP drive? (Keele, Stevenson & Elwert 2020).

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- The model was built to make the **opinion** comparison fair. Nobody asked whether it makes the GDP comparison fair: what drives GDP, and what does GDP drive? (Keele, Stevenson & Elwert 2020).
- The authors agree: "we are not interested here in their statistical and substantive effects" (fn. 69). So does `AGENTS.md`.

8. Assumptions

8.1 What OLS assumes, in plain words

Assumption	**In words**	**If it fails**
Linearity	The average of Y changes by the same amount at every value of X	The line is a poor summary; look at the scatter
Nothing left out	Nothing in the residual is related to X	**The slope is biased** — however many cases
Independent cases	One case tells us nothing about another's residual	Standard errors too small (Module III)
Equal spread	The scatter is equally wide along the line	Usual standard errors off; use robust ones
No exact overlap	No variable is a perfect combination of others	R drops one (`NA`)
Bell-shaped errors	Residuals roughly normal	Matters only in very small samples

Exact overlap is not hypothetical: in Model 4, every country with an Article 98 deal is outside NATO and outside Western Europe, so the paper drops both.

8.2 Two columns again

Identification

Nothing left out.

No amount of data fixes it. More controls help only if they are the right ones. This is the assumption Module IV is about.

Inference

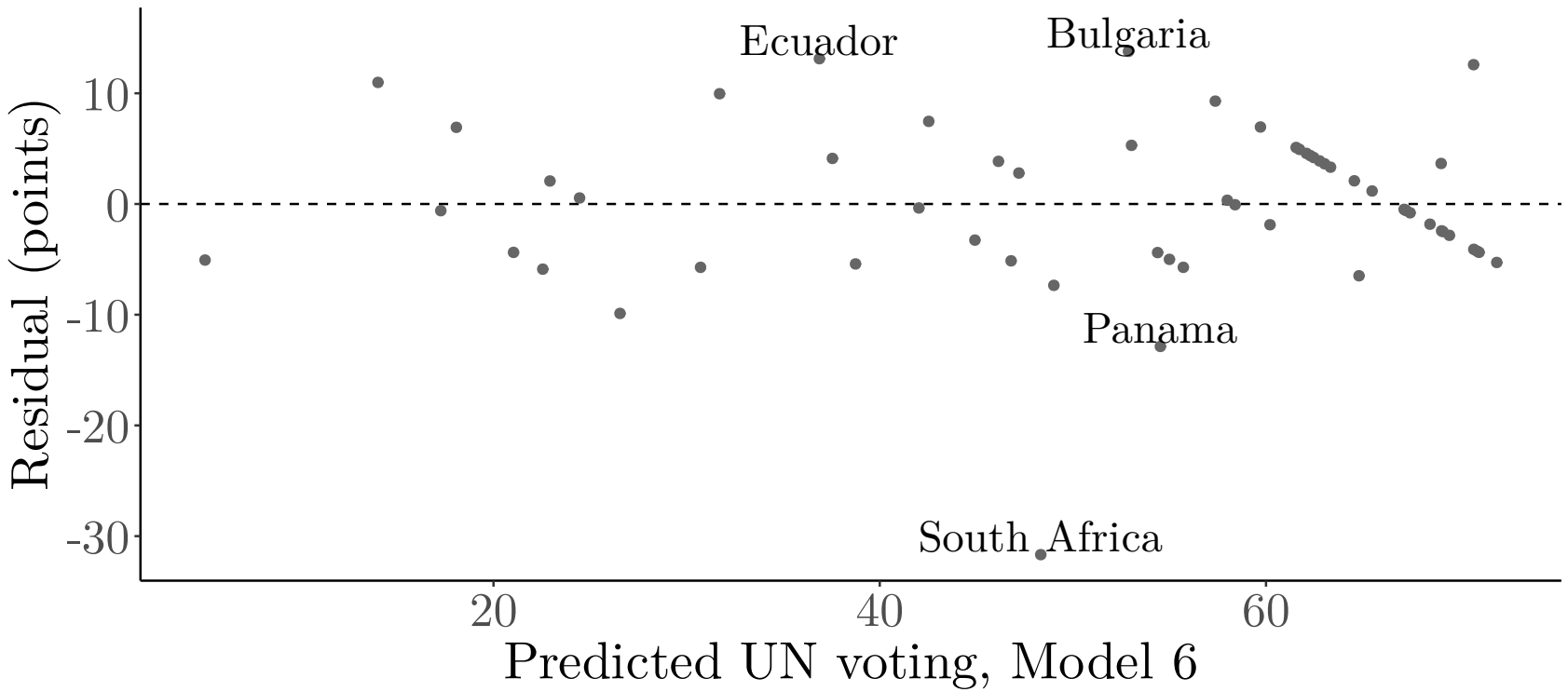
Independence, equal spread, bell shape.

They decide whether the standard errors are honest. Robust standard errors, clustering (Module III), or more data.

A regression output gives you the right-hand column in full: standard errors, t, p. It says nothing about the left-hand column. That part is an argument you make, not a number you print.

9. Diagnostics

9.1 Look at the residuals



What to look for: a curve (linearity), a funnel (unequal spread), a few cases far from the rest. Then find out **who** they are.

9.2 Cases that move the line

Some cases pull harder than others: those **far from the other X values** with **large residuals**.

```
cook <- setNames(cooks.distance(m6), s$country)
round(sort(cook, decreasing = TRUE)[1:3], 2)
```

```
      Georgia South Africa       Nigeria
         0.80         0.18          0.18
```

- Georgia pulls four times harder than any other case (US economic aid: 29 dollars a head, second highest). Drop it: 15.78 becomes 15.49.

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- Albania is the most favourable country by far (0.78; next: 0.39). Drop it: 16.12.

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- Albania is the most favourable country by far (0.78; next: 0.39). Drop it: 16.12.
- Here the result survives. It does not always (Imai's Florida example: one county, Palm Beach).

10. Lab: replicating Table 3

10.1 Lab B: what moves the coefficient?

To the agent

Add to scripts/lab_ols.R. Starting from Model 5, add the eleven controls one at a time (each on its own, not cumulatively) and report the opinion coefficient each time. Then make a residual plot for Model 6 and list the three most influential countries. Run it.

Verify:

1. Which single control moves the opinion coefficient most? Check the change by hand: (control's coefficient) $\times$ (slope from regressing the control on opinion). Does it match?

2. Use the sign table: does the direction of the change make sense, given how that control relates to opinion and to voting?

3. Is the most influential country one you would have guessed? Look up its values. Does dropping it change the conclusion?

4. Which of the eleven controls might be a *consequence* of opinion rather than a cause? Write down your candidate; we start Session 4 with it.

11. What the paper shows

11.1 What would you conclude?

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  With eleven: 15.8 [3.6], far from zero.

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- The authors' own reading: "significant, albeit small" — and smaller than for troops in Iraq, as their salience argument predicts.
- Effect or association? It rests on **nothing left out**. The flip from 2 to 16 is driven mainly by GDP: proof that what we control for matters, and a reason to ask what else might.

11.2 Where this goes

- **Session 4:** the other four models in Table 3. Sent troops to Iraq and signed an Article 98 deal are **yes/no** outcomes. What goes wrong when you draw a straight line through zeros and ones?

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- **Session 4:** the other four models in Table 3. Sent troops to Iraq and signed an Article 98 deal are **yes/no** outcomes. What goes wrong when you draw a straight line through zeros and ones?
- **Module III:** many countries over many years, and an answer to some omitted variables: compare each country **with itself**.
- **Module IV:** when "nothing left out" can be argued by **design** rather than by adding controls.

11.3 For next time

Read:

- Menard (2008), Chapters 1–3
- Kennedy & Waggoner (2021), Chapter 7 (Binary Response Models)
- De Bruin (2018) — we replicate its logistic regressions in Session 4. Come with one question about its research design.
- If today went fast: Dean & Illowsky (2018), Chapter 12; Imai (2017), Chapter 4 — the decoder on the next slide translates their notation

Do:

- Finish Lab B and add a line to `log.md`
- Write down the control you suspect is a consequence of opinion, and why

Session 4: binary outcomes — when the line has to bend.

11.4 Decoder: notation you will meet in the readings

In the readings	What it means on today's slides
$Y = \alpha + \beta X + \varepsilon$	outcome = intercept + slope × predictor + everything else
$\hat{\alpha}$, $\hat{\beta}$, b_0, b_1	the intercept and slope estimated from the data (the "hat" marks an estimate)
$\hat{y}$, $\hat{\varepsilon}$ or e	the predicted value; the residual (observed minus predicted)
$E(Y \mid X)$	the average of Y among cases with a given X
SSR, SSE	sum of squared residuals: what least squares makes as small as possible
r, ρ	the correlation
R^2, RMSE	share of variation accounted for; typical size of a residual
ceteris paribus	"holding the other variables constant"
heteroskedasticity	unequal spread around the line; robust ("HC") standard errors
multicollinearity	controls that overlap heavily (exact overlap: one is dropped)

You do not need to manipulate these. You need to recognise them.